We could split this system in a number of ways. Each of these maps $\exp(t\mathcal{L}_\star)$ may be integrated in the weak sense. Two examples of integrators are the PNHL-S and PNHL-N schemes defined in [228] as

$$e^{h\hat{\mathcal{L}}_{\text{PNHL-S}}} = e^{\frac{h}{2}\mathcal{L}_{\text{A}}} e^{\frac{h}{2}\mathcal{L}_{\text{B}}} e^{\frac{h}{2}\mathcal{L}_{\text{C}}} e^{\frac{h}{2}\mathcal{L}_{\text{D}}} e^{h\mathcal{L}_{\text{O}}} e^{\frac{h}{2}\mathcal{L}_{\text{D}}} e^{\frac{h}{2}\mathcal{L}_{\text{C}}} e^{\frac{h}{2}\mathcal{L}_{\text{B}}} e^{\frac{h}{2}\mathcal{L}_{\text{A}}},$$

and

$$e^{h\hat{\mathcal{L}}_{\text{PNHL-N}}} = e^{\frac{h}{2}\mathcal{L}_{\text{A}}} e^{\frac{h}{2}\mathcal{L}_{\text{B}}} e^{\frac{h}{2}\mathcal{L}_{\text{C}}} e^{\frac{h}{2}\mathcal{L}_{\text{D}}} e^{h\mathcal{L}_{\text{O}}} e^{\frac{h}{2}\mathcal{L}_{\text{D}}} e^{\frac{h}{2}\mathcal{L}_{\text{C}}} e^{\frac{h}{2}\mathcal{L}_{\text{A}}} e^{\frac{h}{2}\mathcal{L}_{\text{B}}}.$$

Traditionally, and elsewhere in this book, we have found symmetric compositions to have superior performance. This is normally due to the fact that they deliver even order of accuracy. In the case of PNHL-S, the supposed advantage would be compounded by the fact that, as in the Verlet and $[\![\text{BAOAB}]\!]$ methods, the forces need only be computed once per timestep. However, rather surprisingly, and for reasons that are as yet unclear, it turns out that the most advantageous technique from the point of view of accuracy with respect to the invariant measure has been observed to be the *nonsymmetric* integration strategy PNHL-N. This method also shows, for a variety of observables, 2nd order approximation of averages. It also allows the use of a much larger integration timestep in practice—so large that it is still more efficient for DPD-like simulations when the extra force calculation is taken into account!

8.10 Numerical Experiments

We investigate a simple test system of 18 unit-mass particles in a 2D periodic 3×3 box using the typical soft DPD potential given in (8.83), with a cutoff radius of 1.0 and $\alpha = 25$. In our simulations the temperature was fixed at $\mathrm{T} = 1$.

We implement the Nosé-Hoover-Langevin and Langevin dynamics schemes, as well as the momentum-conserving PNHL thermostat, and plot the normalized momentum distributions and time evolution of total momentum for the methods in Fig. 8.8. Additionally we compare against the method of Berendsen [32], which modifies Netwon's equations of motion to incorporate a relaxation process for the total system temperature, representing the exchange of energy from the coupling of the external bath. If the instantaneous temperature at time t is $T(t)$, with target temperature T, then using the Berendsen device we have

$$\dot{T}(t) = \gamma_{\mathrm{T}}(\mathrm{T} - T(t)),$$

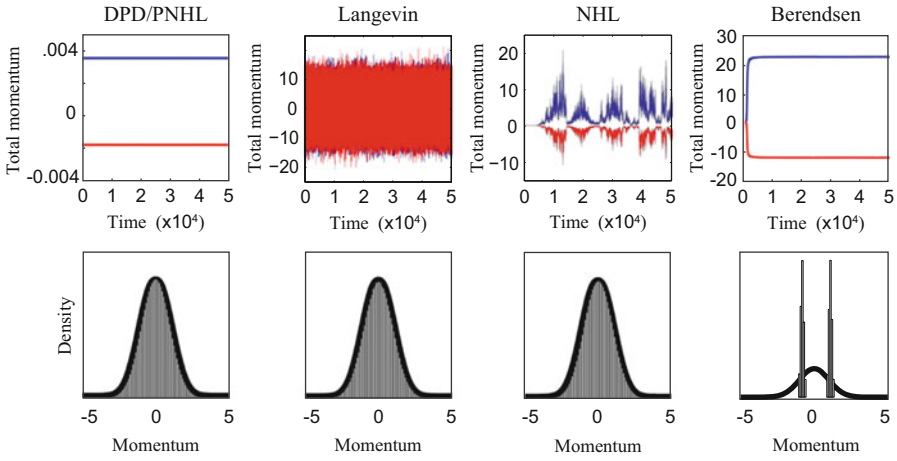


Fig. 8.8 We compare the time evolution of the system's total momentum in the x and y components (*top row*, marked *blue* and *red* respectively), and momentum distributions (*bottom row*) for various methods. In the *lower row* of figures, the *dark*, *solid curve* represents the exact normal density, while the superimposed histograms are constructed from the simulation data. From *left to right*: the PNHL-N thermostat, Langevin dynamics, Nosé-Hoover-Langevin (NHL) dynamics and the Berensen device. The same initial conditions were used with small values of total momenta. The Berendsen device shows a rapid and systematic increase in the momenta, NHL and Langevin exhibit different behavior, obtaining large fluctuating values, whereas the PNHL-N thermostat preserves the total momenta exactly. Standard DPD results are identical to those of PNHL in this respect

for some decay constant $\gamma_{\mathrm{T}} > 0$. The dynamics will not necessarily sample canonically, but the method will ensure that the average temperature is well-controlled (and is hence sometimes useful for equilibration purposes). We implement the version of the scheme given in [186], which completes a step of the velocity Verlet scheme followed by a scaling of the momenta:

$$\boldsymbol{p} \leftarrow \sqrt{1 + \gamma_{\mathrm{T}} h \left(\frac{\mathrm{N_d} k_{\mathrm{B}} \mathrm{T}}{\|\boldsymbol{p}\|^2} - 1 \right)} \boldsymbol{p},$$

for the given timestep h.

We use parameters $\gamma = 4.5$, $\mu = 1$ and $\gamma_{\mathrm{T}} = 1$ for the integrators, with a small stepsize of $h = 0.05$ (making any qualitative sampling errors unlikely). We started the simulations with very small net momenta ($\ell_x = 0.0035$, $\ell_y \approx -0.0019$) and observed a dramatic increase in magnitude of both components when using the simple (deterministic) Berendsen scheme. If the system is ergodic in Euclidean